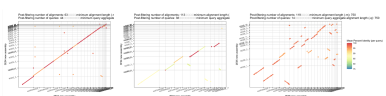
DOINGG@UNION.EDU

COURSE: CSC/BIO243: BIOINFORMATICS

SELF

Contents

Home



The course website for CSC/BIO243, part of the Union College CS curriculum. Here you can find our weekly schedule, assignmnets and resources.

Announcements

Welcome! You may want to bookmark this page for the term.

Office Hours

- Student/Office Hours:
  - Steinmetz 108B
  - Monday 3:00 pm - 4:00 pm
  - Thursday 2:00 pm - 4:00 pm
  - subject to change, check course website for most up-to-date schedule
  - drop-in or schedule a 15 minute slot: https://calendar.app.google/8bus6pfDvyphR9ar5
  - by appointment for another time or over zoom

Course Description

Bioinformatics is the study of how information technology, computer science principles, and algorithmic techniques have affected and informed the study of biology in the 21st century (and vice versa). Specifically, the field of genomics (the study of the function of genes) has generated a tremendous amount of data to be analyzed. Bioinformatics brings to bear data management and analysis techniques found in the information processing field to discover pertinent knowledge in this sea of data that has applications in research and medicine.

In this course, you will learn about both biological and computer science concepts, how they interact with each other, and how they are used together to further research in genomics.

Schedule

Weekly Schedule

Schedule

check often as things may change

W	TOPIC	Reading	Due
1	What is bioinformatics: what are computers and living cells good at?	Bioinformatics for Dummies (BFD) Ch. 1 & 2	Survey, Quiz 0
2	Algorithms & the Genetic Code	Think Python (TP) Ch. 1, 2, 3.4–3.7 & Molecular & Cell Biology (MCFD) Ch. 1, 6, 7	HW 1, Quiz 1
3	Modular Programming & Proteins	TP Ch. 5.1–5.7, 5.11, 6.1 & MCFD Ch. 16, 17	HW 2, Quiz 2
4	Loops & Sequencing	TP Ch. 6.1–6.1, 7.1–7.3, 7.7 & MCFD Ch. 19	HW 3, Quiz 3
5	Sequence Processing & Disease	BFD Ch. 7	HW 4, Quiz 4
6	Sequence Alignment	BFD Ch. 9, TP Ch. 8	HW 5, Quiz 5
7	Multiple Sequence Alignment	BFD Ch. 13	Project 1 Outline

W	TOPIC	Reading	Due
8	Polymerase Chain Reaction	BDF Ch. 13, 9 cont.	Project 1
9	Phylogenetics	BFD Ch. 13 cont.	Project 2 Check-in
10	Gene Prediction	BFD Ch. 5, pp. 145–153	Project 2

Resources

Amoeba Sisters

- videos (and more!) explaining bio topics: https://www.amoebasisters.com/

Python

- Python 3.13.X: the software package needed to write and run Python programs
- https://www.anaconda.com/download/success?reg=skipped

Python Documentation

- Official Documentaion: https://docs.python.org/3/index.html
- Can *always* be used as a reference
- Can be searched for module-specific documentation e.g. turtle

Think Python

- online version of python texbook: https://allendowney.github.io/ThinkPython/
- scroll down to see the interactice google colab notebooks you can *copy to your google drive* and edit for practice

Rosalind Problems

- for python practice: Python Village
- for more bioinformatics problems: Bioinformatics Stronghold
- practice using existing tools (databases, webservers, etc.): Bioinformatics Armory

Runestone (interactive python practice)

- register for an online python practice course (free)
- enter the following as the course name: unioncollege_csc243_spring26
- check out chapters on variables, loops, functions and more python syntax
- includes small practice problems to help you become familiar with useful code
- Runestone has many other open-access online texbooks as well: https://runestone.academy/ns/books/index

Gradescope: used to submit programming assignments

- https://www.gradescope.com/courses/1287277
- Entry Code:G6J44N

Google Colab Pro

- Pro account sign-up: https://colab.research.google.com/signup

- Use your Union email

- A pro account is not needed for this course, just an option Union pays for

- Notes template: https://colab.research.google.com/drive/15rPmKyQKCCbwsdDCpC39o8hczWZ-B5zV?usp=sharing

Python Style Guide

- PEP 8
- should be used in all code cells of Colab notebooks
- Link: https://peps.python.org/pep-0008/

Computer Labs

- Olin 107
- PASTA Lab (ISEC 051+)
- Computer Science Resource room (Steinmetz Hall, 209A)

For help on your homework and projects, available to your are:

- CS Helpdesk (Olin 107, Sun–Thurs 7:00–9:00pm)
- Biology Backup (ISEC 316, Tues & Thurs 5:30–7:30pm)

Grading

Table 2: Grading Scheme

Course Element	Grade Point Contribution	Notes
Participation & Engagement	15%	written answers collected and scanned, 0.5% per activity: itemized list
Quizzes	30%	5 written, 1 oral presentation
Group Projects	40%	coding & written
Final Exam	15%	written
total	100%	

Table 3: Letter Grade Scale

Letter Grade	Percent range
A	93 – 100
A-	90 – 92
B+	87 – 89
B	83 – 86
B-	80 – 82
C+	77 – 79
C	73 – 76

Letter Grade	Percent range
C-	70 – 72
D	60 – 69
F	0 – 59

Assignments

HW 01

CSC/BIO 243: Assignment 1

- Remember to check the schedule for readings: https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html
- Over the weekend
  - finish reading Bioinformatics for Dummies Ch. 1 & 2 and
  - begin the reading Think Python Ch. 1-3 or Molecular & Cell Biology for Dummies Ch. 1,6 and 7
  - you will want to be caught up on the reading since we will have our **first quiz next Wed of Fri**
  - I can answer any questions about materials from readings on Monday, ahead of the quiz

Part 1

- Intro Survey
- Read the Therac25 case study summary and one more detailed article (linked in week1 notes, you can choose one of the two longer articles on the notes page or find another if you prefer. Read critically and carefully and come to class **Wednesday 4/1/26** ready to discuss.

Part 2

- Prepare a 5 minute introduction to a topic for which you bring background knowledge to this class
  - you can use slides, write on the board and/or have me post/print out

handouts

- Each topic will be presented at the beginning of the relevent class day: Topic Sign-up and Schedule
- Based on the results on the survey, I have done my best to match students to topics they are already familiar with, HOWEVER, you should MOVE or ADD your name so you are comfortable with your topic
- Up to 2 students can sign up for each topic
- Questions that arise as you are preparing your topic are great reasons to come to office hours, I am happy to work with you!
- **This presentation will earn you one quiz grade, "Quiz 0" (5% of your final grade)**

HW 02

CSC/BIO 243: Assignment 2 (0.5 pts)

- due Monday 4/20/26 by class time (11:45 am)

~~* due Friday 4/17/26 by class time (11:45 am)~~

~~* due Wednesday 4/15/26 by class time (11:45 am)~~

~~* due Monday 4/13/26 by class time (11:45 am)~~

- Remember to check the schedule for readings: https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html
- Finish the 'Practicing Functions and the Genetic Code' activity that was started in-class as Pair Programming
  - **Bring your written answers to class next ~~Monday~~ ~~Wednesday~~ ~~Friday~~ (0.5 engagement points awarded on these)**
  - You will submit your code, individually, to gradescope
    - submitted code does not hold point values for correctness, but is your **only** chance to receive feedback and start to familiarize yourself with the expectations for graded projects
  - Be sure to add any additional comments (header, description, pseudocode, inline) so the code is a final product rather than a draft
  - You do not need to work on a lab computer (although you always can!), but you should still be working in Idle or a plain text editor, not an Integrated Develeopment Environment (IDE) or a source-code editor with extentions (if this jargon doesn't mean anything to you, you are safe)

- Make sure you are caught up on engagement activities
- Keep checking this page for any updates…

HW 03

CSC/BIO 243: Assignment 3 (0.5 pts)

- due Wednesday 4/22/26 by class time (11:45 am)
- Remember to check the schedule for readings: https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html
- Finish the 'Practicing Conditionals, Loops and Protein Tranlsation' activity that was started in-class as Pair Programming
  - **Bring your written answers to class next Wednesday (0.5 engagement points awarded on these)**
  - You will submit your code, individually, to gradescope
    - submitted code does not hold point values for correctness, but is your **only** chance to receive feedback and start to familiarize yourself with the expectations for graded projects
  - Be sure to add any additional comments (header, description, pseudocode, inline) so the code is a final product rather than a draft
  - You do not need to work on a lab computer (although you always can!), but you should still be working in Idle or a plain text editor, not an Integrated Develeopment Environment (IDE) or a source-code editor with extentions (if this jargon doesn't mean anything to you, you are safe)
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Weekly Notes

Week 1

Notes from Week 1

For Reading/HW

- Therac25 Summary
- Therac25 Leveson
- Therac25 Silvis

Monday in class

- No More DNA

Wednesday in class

- Slides
- What are computers good at?

Friday in class

- Anatomy of a function
- Logging on to lab computers
- Pair Programming
- Rosalind Problems for Learning Python
- Rosalind Bioinformatics Practice Problems
- Rosalind Bioinformatics Textbook Problems

Week 2

- Mon:
  - Therac25 discussion (see notes from last week)
  - Therac25 in class prompt
  - Pseudocode practice activity
- Wed: Practicing Functions and the Genetic Code
  - note, we will start the above activity in class on Friday
  - demo code of comments and functions
  - demo code to make random pairs of bio-cs names
  - presentation on Iteration and note taking activity
- Fri: we will have a quiz on material up through Monday
  - we will talk more about what DNA is and how it is turned into protein and/or replicated
  - we will also begin the assignment above as a pair programming activity
  - presentation on DNA replication and note taking activity

Week 3

- Mon: Practicing Functions and the Genetic Code
  - note, the above activity was handed out in class last Friday
  - demo code of comments and functions
  - slides from week 2 on introductions to the central dogma and the genetic code
  - slides from student presentations
  - presentation on Conditionalsand note taking activity
- Wed: Coding Warm-up
  - Practicing Conditionals, Loops and Protein Tranlsation
  - algorithm design reference
  - pseudocode translation reference
  - presentation on Function Design and note taking activity
- Fri: Quiz #2
  - DNA replication, transcription, translation
  - functions, loops, conditionals
  - pseudocode, algorithm design
  - presentation on Protein Synthesis and note taking activity

About

References